

Vulnerability Assessment Report

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business? The employees regularly query the server to find potential customers.*
- *Why is it important for the business to secure the data on the server? This is how the business grows and gets new clientele.*
- *How might the server impact the business if it were disabled? It will put a pause on a big source of revenue for the business.*

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Obtain sensitive information via exfiltration, perform reconnaissance and surveillance of organization- threat source examines and assesses the company's vulnerabilities over time using various tools (e.g., scanning, physical observation).	1	3	1
Customer	Might alter data unintentionally.	2	2	3

<i>Hacker</i>	<i>Can use access to public info to gain sensitive information about the system.</i>	<i>1</i>	<i>3</i>	<i>2</i>
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Approach

After careful consideration, several threat sources present potential risks to the business, including competitors, customers, and external hackers. Due to the open nature of the server, these threat actors could potentially access sensitive information, unintentionally alter data, or misuse public or confidential information for malicious purposes.

Employees regularly rely on data from this server to acquire new customers. Any disruption to this information stream could directly impact business operations and result in a loss of revenue. Additionally, the deletion or compromise of critical data could threaten day-to-day systems that are essential to business continuity.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database, to the employees that are in different locations, using a VPN would provide an encrypted tunnel straight to the database, protecting the data even further.